Ex.No: 18

Date:

Aim:

To implement Cursors using a program in MySQL.

Description:

Cursor is a temporary memory or temporary work station used by the database server while performing DML (Data Manipulation Language) operations. Cursors are used to process table data.

1. Implicit Cursors:

Implicit Cursors are default cursors allocated automatically by the database system when DML operations are performed.

2. Explicit Cursors:

Explicit Cursors are created by users when required. They are mainly used for fetching data from a table in a row-by-row manner.

Steps to create an Explicit Cursor:

1. Declare Cursor Object.

Syntax: DECLARE cursor_name CURSOR FOR SELECT * FROM table_name;

2. Open Cursor Connection.

Syntax: OPEN cursor_name;

3. Fetch Data from Cursor.

The following methods can be used to access data from a cursor: FIRST, LAST, NEXT, PRIOR, ABSOLUTE n and RELATIVE n.

FIRST – Fetches the first row from the cursor.

LAST – Fetches the last row from the cursor.

NEXT – Fetches the next row in the forward direction.

PRIOR – Fetches the previous row in the backward direction.

ABSOLUTE n – Fetches the nth row.

RELATIVE n – Fetches data relative to the current row.

4. Close Cursor Connection.

Syntax: CLOSE cursor_name;

5. Deallocate Cursor Memory.

Syntax: DEALLOCATE cursor_name;

Problem-1:

Write a Cursor program using MySQL to retrieve the email-ids (build an email list) of employees from the employees table.

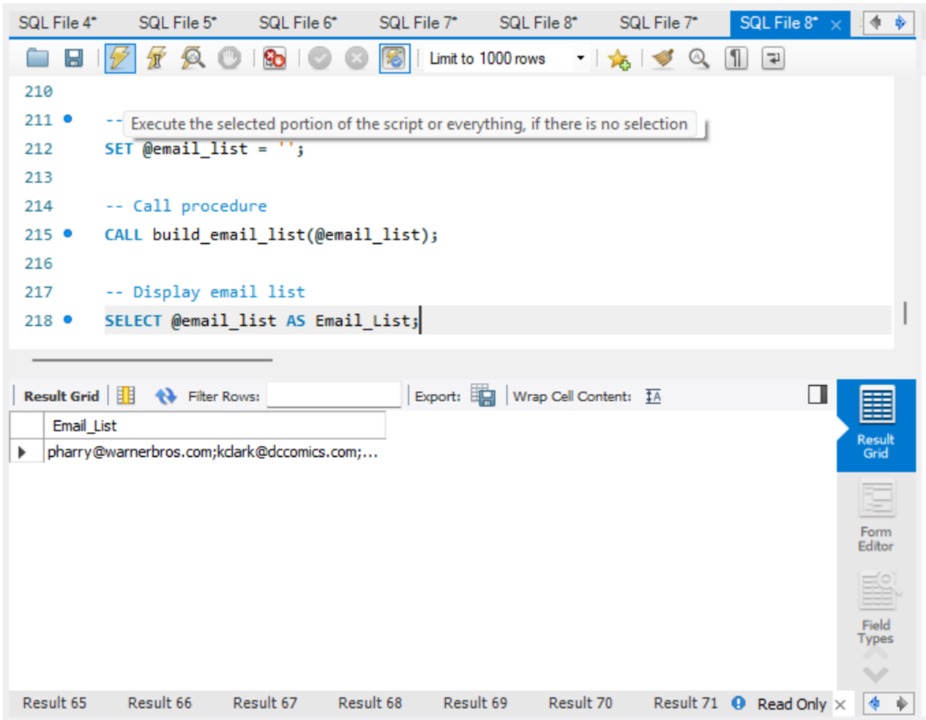
Solution:

```
CREATE TABLE employees(
   id INTEGER,
   Name VARCHAR(100),
   email VARCHAR(100)
);

INSERT INTO employees(id, Name, email)
VALUES (1, 'Harry Potter', 'pharry@warnerbros.com');

INSERT INTO employees(id, Name, email)
VALUES (2, 'Clark Kent', 'kclark@dccomics.com');

INSERT INTO employees(id, Name, email)
VALUES (3, 'Tony Stark', 'stony@marvel.com');

DELIMITER $$

CREATE PROCEDURE build_email_list(INOUT email_list VARCHAR(4000))
BEGIN
   DECLARE v_finished INTEGER DEFAULT 0;
   DECLARE v_email VARCHAR(100) DEFAULT '';

   -- Declare cursor for employee email
   DECLARE email_cursor CURSOR FOR
      SELECT email FROM employees;
```

```
    -- Declare NOT FOUND handler
    DECLARE CONTINUE HANDLER FOR NOT FOUND
        SET v_finished = 1;

    OPEN email_cursor;

    get_email: LOOP
        FETCH email_cursor INTO v_email;

        IF v_finished = 1 THEN
            LEAVE get_email;
        END IF;

        -- Build email list
        SET email_list = CONCAT(v_email, ';', email_list);
    END LOOP get_email;

    CLOSE email_cursor;
END$$

DELIMITER ;

-- Calling the procedure and getting the email list
SET @email_list = '';

CALL build_email_list(@email_list);

SELECT @email_list;
```

Expected Output:

The SELECT statement displays the email IDs as a semicolon-separated email list. The order may appear in reverse because each fetched email is added to the beginning of the list.



Result:

Thus, the Cursor program using MySQL was executed successfully and the email IDs of employees were retrieved using an explicit cursor.